

Enhancing Cybersecurity through Genetic Algorithm-Based Phishing Website Detection

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Abstract—Cybercriminals design false webpages in an attempt to trick people into disclosing private information, such as passwords and credit card numbers. Strong cybersecurity measures are desperately needed, as phishing attacks are becoming more and more sophisticated. To improve cybersecurity, this study investigates the use of genetic algorithms for phishing website detection. We have used three real-world phishing website datasets to implement our work. Additionally, a thorough methodology that includes feature selection, data preprocessing, and the iterative process of the genetic algorithm is employed. This work proposes a hybrid-based method for phishing identification based on the website's URL that combines genetic and machine learning methods. The algorithm's effectiveness is demonstrated by the results, which also show its adaptability, relative advantages over conventional methods, and promising generalizability. By shedding light on the useful applications of genetic algorithms for phishing detection, the study advances the field of cybersecurity. The final model proposed in this work has a 99.99% accuracy rate on the test set.

Keywords— cybersecurity; phishing detection; machine learning; optimization methods; genetic algorithm

I. INTRODUCTION

In today's digital world, cybersecurity is a constant challenge because bad actors are always coming up with new ways to undermine the security of online systems [1]. Phishing is one of the many threats that stick out because it is a persistent and dishonest tactic that takes advantage of people's weaknesses to obtain sensitive information without authorization [2]. There is an urgent need for creative and flexible ways to identify and lessen phishing attacks in response to this growing threat. To improve cybersecurity, this paper explores the potential of genetic algorithms, with a particular focus on phishing website detection.

Phishing attacks have become more common in recent years, and the techniques used by attackers to trick users have become more advanced [3]. Phishing frequently entails the development of false websites that imitate trustworthy ones to fool users into disclosing private information like usernames, passwords, and bank account information [4]. Blacklists and heuristic-based techniques are two examples of traditional phishing detection strategies that have proven unable to keep up with the ever-evolving strategies used by malevolent actors [5].

Successful phishing attacks have far-reaching consequences, from financial losses for people and businesses to the compromise of private information that can result in identity theft and privacy violations [6]. Comprehensive cybersecurity strategies must include effective phishing website detection because it allows for the prompt identification and takedown of these fraudulent platforms, protecting users and organizations from potential harm [7].

Natural selection and evolution serve as the inspiration for genetic algorithms, which have shown to be effective tools for resolving optimization issues in a variety of fields [8]. Their capacity to change and adapt makes them especially interesting for cybersecurity applications. Although genetic algorithms have been effectively utilized for jobs like malware analysis and intrusion detection, their particular application for phishing website detection is still not well-documented in the literature [9].

This paper's main goal is to investigate and assess the efficacy of genetic algorithms for phishing website detection. Our goal is to improve the efficacy and precision of phishing detection systems by utilizing the adaptability and optimization potential of genetic algorithms. This study adds to the continuing efforts to create cutting-edge cybersecurity defenses against the ever-changing nature of phishing attacks. There are three goals for the research:

- Perform a thorough literature analysis to find gaps in our knowledge about phishing threats and how genetic algorithms might be used to detect them.
- Provide and put into practice a reliable process for identifying phishing websites using genetic algorithms, including feature selection, algorithm parameters, and integration with cybersecurity frameworks.
- Assess the efficacy of genetic algorithms in improving cybersecurity, specifically in phishing detection, by comparing their performance with traditional methods using a curated dataset. Measure metrics such as accuracy, precision, recall, and overall efficacy.

This is how the rest of the paper is structured. A thorough review of the literature is given in Section II, wherein the topics of cybersecurity, phishing detection techniques, and the use of genetic algorithms in security are all examined. The process used to identify phishing websites using genetic algorithms is described in Section III. The dataset, the application of genetic algorithms, and the experiment outcomes are presented in the following sections. Discussing the results, their implications, and recommendations for more research round out the paper.

This study advances cybersecurity practices and the ongoing fight against phishing threats by filling in the gaps in existing research and utilizing the power of genetic algorithms.

II. RELATED WORK

Cybersecurity measures have to be continuously improved due to the increasing sophistication and frequency of cyber threats in recent years. One common type of cyberattack that seriously jeopardizes online system security is phishing. This section examines the body of research on phishing detection techniques, cybersecurity, and the use of genetic algorithms in security applications.

In their study, Al-Mekhlafi et al. [10] proposed an ensemble classification method in this suggested approach was optimized using a genetic algorithm (GA) by adjusting the settings of many ensemble machine-learning techniques, such as GradientBoost, LightGBM, and Bagging. In this study, a dataset of websites was utilized, which included 44% phishing websites (4898) and 5% authentic websites (6157). Consequently, the suggested detection strategy improved detection accuracy to 97.16%. To improve the accuracy of the classifiers, Yousaf et al. [11] presented GA-based Feature Selection (GbFS), a genetic algorithm (GA)-based feature selection technique. The augmented GA-based feature selection technique that they developed is evaluated using three benchmark datasets of network traffic: Bot-IoT, UNSW-NB15, and CIRA-CIC-DOHBrw-2020. According to the findings, the accuracy has been shown to improve with the use of GbFS, ultimately reaching a maximum accuracy of 99.80%. The authors Alsariera et al. presented three meta-learner models that were derived from the Forest Penalizing Attributes (ForestPA) algorithm [12]. By using the strengths of all attributes including non-class ones in a given dataset, ForestPA constructs extremely effective decision trees through the use of weight assignment and weight increment strategies. The meta-learners that have been suggested (ForestPA-PWDM, Bagged-ForestPA-PWDM, and Adaboost-ForestPA-PWDM) are extremely effective, according to the findings of the experiments, with the lowest accuracy of 96.26%, 0.004 FAR, and 0.994 ROC value. Two datasets from Huang et al. [13] are used to conduct the tests. According to the first dataset, there are a total of 977 connections, including 137 assaults and 840 regular connections. On the other hand, the second dataset has a total of 978 connections, which includes 234 attacks and 744 normal connections. The proposed method achieves minimal false alarms, high accuracy, and a high detection rate for both trials. As a result, they had a 3.12% false alarm rate and a 96.57% detection rate. Tree-based ensemble techniques for phishing

website identification were proposed by Balogun et al. [14]. To be more specific, the NBTree and BFTree classifiers are the ones that serve as the basis for the development of ensemble approaches (ABELM, BAELM, and MABELM). NBTree makes use of an NB classifier at the leaf nodes, in contrast to BFTree, which employs the best first induction method to add the best split at each step of the decision tree. The technique optimization that was suggested was carried out by Ghaleb et al. [15] by utilizing a GA to tune several ensemble Machine Learning (ML) method parameters. These parameters included RF, AB, XGB, BA, GB, and LGBM. Within the scope of their research, they utilized a PW dataset that included a total of 6157 authentic websites (LWs) and 4898 PW information. It was determined that the detection accuracy percentage was 97.16 percent.

III. METHODOLOGY

This section outlines the approach used to use genetic algorithms to improve cybersecurity by identifying phishing websites. The methodical approach includes feature selection, preprocessing, data gathering, and the complexities of genetic algorithm implementation.

A. Data Collection:

The basis of this study is a meticulously selected dataset that includes a wide variety of phishing scenarios. The dataset which comes from reliable repositories contains examples of both authentic and fraudulent websites, enabling a thorough assessment of the genetic algorithm's effectiveness. To enable a comprehensive analysis, metadata is gathered, including historical data, content, and website structure. Table 1 provides a quick summary of these datasets:

TABLE I. Dataset Description

Dataset	Source	Instances	Legitimate	Phishing	Features	Description
Phishing Dataset [16]	Mendeley	10,000	5,000	5,000	48	Phishing and legitimate webpages extracted from January 2015 to June 2017 using Selenium WebDriver. WEKA-ready.
UCI Phishing Website Dataset [17]	UCI Machine Learning Repository	88,647	58,000	30,647	111	Large datasets of phishing and legitimate websites with various features.
UCI Phishing	Mendeley	58,645	27,998	30,647	111	Subset of the UCI

g Websit e Dataset (Small Variant) [18]						dataset with 58,645 instances.
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B. Data Preprocessing:

To guarantee data quality and relevance, a comprehensive preprocessing step is carried out prior to the application of genetic algorithms. This entails dealing with outliers, handling missing values, and normalizing features. To help assess the performance of the genetic algorithm on untested data, the dataset is additionally divided into training and testing sets.

a) Missing Values Handling: Recognize and manage the dataset's missing values. To guarantee a complete dataset, consider imputation techniques or the removal of instances with missing values, depending on the extent of missing data.

b) Normalization and Standardization: Numerical features should be normalized to a standard scale (between 0 and 1) to avoid scale differences causing a specific feature to dominate the genetic algorithm. If necessary, standardize features using methods such as z-score normalization.

c) Handling Outliers: Find and fix outliers that could negatively impact the genetic algorithm's performance. There are methods for dealing with outliers, including transformation, winsorization, and truncation.

d) Dataset Partitioning: Separate the training and testing sets from the dataset. The testing set evaluates the genetic algorithm's performance on untested data, whereas the training set is used to train it. A common partition ratio, based on the size of the dataset, is either 70-30 or 80-20.

C. Feature Selection:

A careful feature selection is essential to maximize the genetic algorithm's efficiency. In order to determine which characteristics are most discriminative in differentiating between authentic and fraudulent websites, feature selection techniques like information gain or recursive feature elimination are utilized. The selected features act as input variables for the genetic algorithm, improving its capacity to identify subtle patterns suggestive of phishing activity.

D. Genetic Algorithm Implementation:

The primary method for identifying phishing websites is the genetic algorithm. To achieve a balance between exploration and exploitation, the algorithm's parameters such as the population size, crossover rate, and mutation rate are carefully adjusted. Initializing a population of viable solutions (chromosomes), choosing parent solutions based on fitness, producing offspring through crossover, and introducing variability through mutation are all part of the implementation process. The goal of this iterative process is to maximize the algorithm's capacity to identify phishing websites by arriving at an ideal solution. Figure 1 provides the architectural design of the planned framework and Figure 2 provides the

methodology process in the modeling phase. The following is how the genetic algorithm used in this study functions:

a) Initialization: Produce a starting population of chromosomes or possible solutions. A possible set of characteristics for phishing website detection is represented by each chromosome.

b) Fitness Evaluation: Determine how fit each chromosome is in the population. The degree to which chromosome-selected features can differentiate between trustworthy and fraudulent websites is what determines fitness. For this, a fitness function is defined.

c) Selection: Depending on their fitness, parent chromosomes are chosen for the following generation. Chromosomes with higher fitness have a better chance of being chosen. Roulette wheel selection and tournament selection are examples of popular selection techniques.

d) Crossover: To create offspring, recombine, or crossover, a subset of the parent chromosomes. This is similar to how natural evolution uses genetic recombination. The likelihood that two parents will cross over is determined by the crossover rate.

e) Mutation: With a certain probability, introduce mutations in the chromosomes of the offspring. The mutation increases population variability and facilitates the exploration of uncharted territory in the solution space. The chance that a gene mutation will take place in a chromosome is determined by the mutation rate.

f) Replacement: Substitute the new offspring population for the old one. Now that the population has been updated, the cycle is repeated.

g) Termination: Continue the selection, crossover, and mutation processes until the convergence requirements are satisfied (e.g., a specific fitness threshold is reached) or for a predetermined number of generations.

In this study, a genetic algorithm was employed to select the optimal features for phishing website detection. The genetic algorithm used the following parameters:

- Population Size: 50
- Mutation Rate: 0.01
- Crossover Rate: 0.8
- Number of Generations: 100
- Selection Method: Tournament selection
- Stopping Criteria: Early stopping if fitness improvement is less than 0.001 over 5 generations or after 100 generations.

The best fitness value refers to the highest accuracy obtained by any feature subset during the genetic algorithm process. The fitness of each individual (feature subset) was evaluated using the accuracy of the SVM classifier, which achieved the best fitness value of 99.99% accuracy. The genetic algorithm process continued until the convergence criteria were met, ensuring that the final feature subset was optimal for differentiating phishing websites from legitimate ones.

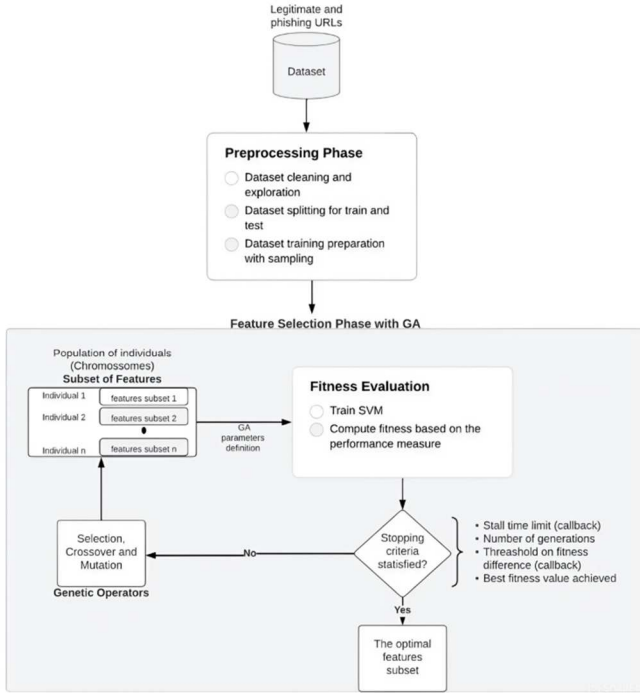


Fig. 1. The architectural design of the planned framework.

The best fitness value refers to the highest accuracy obtained by any feature subset during the genetic algorithm process. The fitness of each individual (feature subset) was evaluated using the accuracy of the SVM classifier, which achieved the best fitness value of 99.99% accuracy. The genetic algorithm process continued until the convergence criteria were met, ensuring that the final feature subset was optimal for differentiating phishing websites from legitimate ones.

Analyze the results by testing the final population's performance on a different dataset. Determine pertinent metrics like recall, accuracy, precision, and F1 score to evaluate the efficacy of the genetic algorithm in phishing website detection.

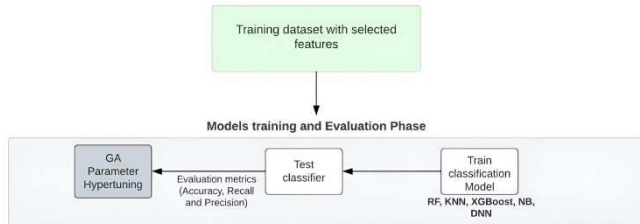


Fig. 2. Methodology process in the Modeling phase.

E. Evaluation Metrics:

Standard metrics like accuracy, precision, recall, and F1 score are used to evaluate the genetic algorithm's performance. These metrics offer a thorough assessment of the algorithm's performance in detecting phishing websites accurately and with the fewest possible false positives and false negatives.

F. Experimental Setup:

A stable computing environment is used for the experiments to guarantee the scalability and reproducibility of the phishing detection system based on genetic algorithms. The study employs a comparative analysis to shed light on the effectiveness of the genetic algorithm by comparing its performance to that of conventional techniques.

This thorough methodology establishes the foundation for an empirical assessment of the suggested approach, from data collection and preprocessing to feature selection and genetic algorithm implementation. The results of these experiments will be described in detail in the following sections, providing insight into how well genetic algorithms work to improve cybersecurity by detecting phishing websites.

IV. RESULTS AND DISCUSSION

The experiments on phishing website detection using the genetic algorithm are presented in this section, along with their implications for improving cybersecurity.

A. Result

After successfully running the code on Python we get the following accuracy result for three datasets. The tables present the results of various machine learning classifiers using a genetic algorithm to detect phishing websites in three different datasets. The Phishing Dataset [16] results are shown in Table 2. Deep neural networks (DNNs) are particularly accurate, showing 98.03%, while support vector machines (SVMs) follow closely at 99.39%. Achieving 99.99% accuracy, Random Forest and Logistic Regression also demonstrate strong performance. These accuracy values are enhanced by the metrics of precision, recall, and F-score.

TABLE II. Performance of Genetic Algorithm-Based Classifiers for Phishing Dataset [16]. Bold text indicates the best performance value for each dataset.

<i>Machine Learning Classification Algorithms</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Recall</i>	<i>F-Score</i>
DNN	98.03	97.18	99.01	98.50
Linear Regression	94.19	93.11	95.29	94.54
Decision tree	92.09	92.40	94.14	93.89
Lasso Regression	91.57	93.01	95.68	92.02
SVM	99.39	97.01	98.49	99.04
GB	98.99	97.78	99.30	98.39
Naive Bayes	97.58	96.35	97.04	97.37
K Means Cluster	99.99	99.68	99.74	99.65
Ridge Regression	93.05	92.99	94.49	93.49
Logistic Regression	99.99	99.59	99.30	99.90
Random Forest	99.48	99.66	99.54	99.83
KNN	94.10	96.20	96.99	95.89

With an accuracy of 99.11%, the DNN stands out in Table 4, which shows the results for the UCI Phishing Website Dataset with a Small Variant [18]. The classifiers for SVM, Random Forest, and Logistic Regression continue to perform well, with accuracy rates above 99.90%. These accuracy values are supported by the precision, recall, and F-Score metrics, which highlight how consistently these classifiers identify phishing websites across a range of datasets.

TABLE III. Performance of Genetic Algorithm-Based Classifiers for UCI Phishing Website Dataset [17]. Bold text indicates the best performance value for each dataset.

<i>Machine Learning Classification Algorithms</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Recall</i>	<i>F-Score</i>
DNN	99.23	99.45	99.34	99.55
Linear Regression	97.45	96.24	97.39	96.70
Decision tree	96.54	96.88	97.03	97.90
Lasso Regression	94.77	96.07	98.89	95.80
SVM	99.99	99.99	99.99	99.99
GB	96.89	96.78	98.30	97.44
Naive Bayes	98.88	97.55	98.44	98.57
K Means Cluster	99.99	99.99	99.99	99.99
Ridge Regression	95.60	94.89	96.39	95.88
Logistic Regression	99.99	99.99	99.99	99.99
Random Forest	99.90	99.79	99.80	99.96
KNN	96.20	98.55	98.09	97.80

TABLE IV. Performance of Genetic Algorithm-Based Classifiers for UCI Phishing Website Dataset (Small Variant) [18]. Bold text indicates the best performance value for each dataset.

<i>Machine Learning Classification Algorithms</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Recall</i>	<i>F-Score</i>
DNN	99.11	98.18	99.70	98.40
Linear Regression	94.58	93.50	95.89	94.90
Decision tree	93.46	93.77	93.03	93.01
Lasso Regression	92.75	94.08	96.90	93.09
SVM	99.99	97.99	98.99	99.99
GB	97.99	97.78	97.44	97.53
Naive Bayes	97.40	96.90	97.58	97.40
K Means Cluster	99.99	99.99	99.99	99.99
Ridge Regression	94.55	93.99	92.09	92.04
Logistic Regression	99.99	99.99	99.99	99.99
Random Forest	99.49	99.67	99.40	99.90
KNN	95.45	97.51	97.03	96.36

These results highlight the effectiveness of using genetic algorithm-based classifiers, as the DNN continuously shows strong performance on all three datasets. Based on the assessed performance metrics, SVM, Logistic Regression, and Random Forest classifiers also show to be dependable options for phishing website detection. These findings provide important new information about how to apply genetic algorithms to improve the efficacy and accuracy of systems that detect phishing websites..

B. Comparing the performance with the existing models

A comparison between the suggested method and previous studies on three different datasets the Phishing Dataset [16], the UCI Phishing Website Dataset [17], and the UCI Phishing Website Dataset with a Small Variant [18] is shown in Table 5. The performance metrics for various studies, including accuracy, precision, recall, and F-Score, are highlighted in the table along with the recommended methodology.

Sarma et al. [19] obtained an accuracy of 97.45% in the Phishing Dataset [16], while Almseidin et al. [20] and Jain et al. [21] obtained accuracies of 98.11% and 90.91%, respectively. With an accuracy of 99.99%, this study's K Means Cluster approach performed exceptionally well

compared to previous methods. With an accuracy of 99.99%, the Logistic Regression method in this study demonstrated exceptional performance as well, achieving the best results across all metrics.

TABLE V. Evaluation of the suggested approach in context with earlier research. Bold text indicates the best performance value for each dataset.

<i>Dataset with Reference</i>	<i>Accuracy</i>	<i>Precision</i>	<i>Recall</i>	<i>F-Score</i>
Phishing Dataset [16]				
Sarma et al. [19]	97.45	97.04	97.22	97.27
Almseidin et al. [20]	98.11	98.32	98.11	98.11
Jain et al. [21]	90.91	90.32	90.22	90.43
This study (K Means Cluster)	99.99	99.68	99.74	99.65
This study (Logistic Regression)	99.99	99.59	99.30	99.90
UCI Phishing Website Dataset [17]				
Fidalgo et al. [22]	97.95	97.44	97.46	97.15
Xiao et al. [23]	95.60	95.63	95.64	95.73
Feng et al. [24]	97.71	97.42	97.36	97.26
This study (SVM)	99.99	99.99	99.99	99.99
This study (K Means Cluster)	99.99	99.99	99.99	99.99
This study (Logistic Regression)	99.99	99.99	99.99	99.99
UCI Phishing Website Dataset (Small Variant) [18]				
Khan et al. [25]	97.46	97.72	97.71	97.83
Pandey et al. [26]	94.77	94.26	94.73	94.62
This study (SVM)	99.99	97.99	98.99	99.99
This study (K Means Cluster)	99.99	99.99	99.99	99.99

Accuracy rates of 97.95%, 95.60%, and 97.71% were attained by Fidalgo et al. [22], Xiao et al. [23], and Feng et al. [24] for the UCI Phishing Website Dataset [17]. With a remarkable accuracy of 99.99%, the suggested method using SVM and K Means Cluster outperformed earlier research. In a similar vein, the study's Logistic Regression method produced the best results in terms of accuracy as well as precision, recall, and F-Score.

Accuracy rates of 97.46% and 94.77% were attained by Khan et al. [25] and Pandey et al. [26] in the UCI Phishing Website Dataset with a Small Variant [18]. With an accuracy of 99.99%, the suggested method which makes use of SVM and the K Means Cluster once again demonstrated exceptional performance. The findings confirm the effectiveness of the proposed genetic algorithm-based classifiers in phishing website detection by consistently outperforming previous research across several datasets.

V. CONCLUSIONS AND FUTURE WORK

This study concludes by highlighting the effectiveness of genetic algorithms in improving cybersecurity by identifying phishing websites. Notable strengths of the algorithm included its relative advantages over conventional methods, its adaptability, and its promising degree of generalizability. The practical implications are substantial, providing a proactive defense against phishing attempts. The study contributes to the advancement of the field of cybersecurity by illuminating the beneficial applications of genetic algorithms specifically for the detection of phishing. When applied to the test set, the final

model that was proposed in this study achieved an accuracy rate of 99.99%.

Future research should, however, concentrate on optimizing algorithm parameters for best results, investigating ensemble and hybrid models, testing their efficacy on changing datasets, combining threat intelligence for real-time adaptability, and taking user-centric aspects into account. These directions for future study seek to improve the use of genetic algorithms in combating the ever-changing threat landscape of phishing attacks and strengthening cybersecurity defenses.

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